Vijayalakshmi (Viji) Ramasamy, Ph.D.
Assistant Professor (Tenure Track), Computer Science, AEP College of Engineering and Computing
Georgia Southern University, Statesboro Campus, USA
Email: vramasamy@georgiasouthern.edu

Education

- **Ph.D. Computer Science**, Anna University, Chennai, India **2013**
  Dissertation: *Efficient Algorithms and Data Structures for Storage and Mining in Large Graph Databases*
- **M.Phil.** Master of Philosophy in Computer Science, Bharathiar University, Coimbatore, India **2003**
  Thesis: *Methodology to Evaluate Code-based Regression Test Selection Techniques and Predictors of Cost Effectiveness*
- **M.C.A**. Master's in Computer Applications, Bharathiar University, Coimbatore, India **1996**
- **D.D.B.M.** Diploma in Database Management, Info-Drive Software Solutions, Coimbatore, India **1996**
- **B.Sc.** Chemistry, Bharathiar University, Coimbatore, India **1993**

Academic Appointments

- **Assistant Professor (Tenure Track)**, Department of Computer Science, AEP College of Engineering and Computing, Georgia Southern University, GA, USA — 2024 – present
- **Tenured Associate Professor**, Department of Computer Science, University of Wisconsin-Parkside (UW-Parkside), WI, USA Tenured Associate Professor-UW-Parkside_Link — 2023 – 2024
- **Associate Professor,** Department of Computer Science, UW-Parkside, WI, USA — 2020 – 2024
- **Visiting Assistant Professor,** Department of Computer Science and Software Engineering, Miami University (MU), Ohio, USA — 2016 – 2020
- **Associate Professor,** Department of Applied Mathematics and Computational Sciences, PSG College of Technology, Anna University (AU), India — 2013 – 2016
- **Assistant Professor,** Department of Applied Mathematics and Computational Sciences, PSG College of Technology, Anna University (AU), India — 2010 – 2013
- **Senior Lecturer,** Department of Computer Applications, PSG College of Technology, Anna University (AU), India — 2004 – 2010
- **Lecturer,** Department of Computer Science, Bharathiar University, India — 1997 – 2004
- **Jr. Systems Analyst,** Maxwell Industries Ltd, India — 1996 – 1997

Professional, Administrative, and Special Appointments

- **Chair, Academic Director of Master of Science Computer Information Systems (MSCIS)**, Department of Computer Science, UW-Parkside, WI, USA — 2023 – 2024
- **Adjunct Faculty,** Cognitive Neuro-Engineering Laboratory, University of South Australia (UniSA), Australia — 2013 - 2016
- **Principal Investigator and Research Fellow:** Computational Neuroscience Laboratory (CNSL), PSG College of Technology, Anna University (AU), India — 2012 - 2019

Research Areas

- **Data Science & Graph Analytics:** scalable algorithms for large graphs; Bayesian & social/collaboration networks
- **AI in Education (AIED):** Eye-tracking–based analytics to quantify student attention and AI-enabled Learning Assistance for learning engagement in computing education.
- **ML/NLP in Software Engineering:** user stories & requirements, software text understanding
- **Computational Neuroscience:** EEG-based functional brain networks; connectivity during cognition
- **Privacy & Policy Analytics:** transparency/compliance modeling using NLP and graph mining in search warrant analysis, privacy policy analysis,
- **Cybersecurity Analytics:** cybersecurity knowledge graphs for threat investigation and pattern mining; linkage to digital forensics text and code artifacts
- **Integrative Environmental Health & Cognition:** AI/ML to model environmental/dietary impacts on cognitive processes and health

Research Grants & Sponsored Projects

Active/Ongoing

- **NSF CNS - 2449818 - $219,994, PI, 2023-2027**, *Collaborative Research: CUE-T: Broadening Participation in Computing via Active Learning Strategies in Multi-Institution Online Synchronous Learning Environments* (**$2M Award,** FIU, AAMU, FAMU, HSU, and RU).
- **Affordable Materials Grant, Round 27 - Transformation Grants –$10,566.44, Co-PI, Spring 2025-Spring 2026,** Student Participation in Materials Evaluation and/or Development, Course: *CSCI 7090 Data Science and Machine Learning Project*.
- **PCEC Undergraduate Research** - Faculty Advisor, Spring 2026
  - (i) **Recipe for Code: Immersive Java Gaming, $1500** - Success Emmanuel, Sean Roe, Naomi Oropesa - Open-source game development for Java Programming
  - (ii) **AI Navigator: Interactive Game Development for AI Literacy, $1500** - Sean Roe, Naomi Oropesa, Success Emmanuel - Open-Source game for AI Literacy training

Submitted

- **NIH-PA-25-246-R03 - Co-PI, $283,676, (2026-2028**), *Low-Cost Graph Learning Approaches to Advance Early Prediction and Generalizability in Alzheimer's Disease and Related Dementias*, October 15, 2025.
- **EPA EE - Co-PI, $275,135, (Nov. 1, 2026 – Oct. 31, 2028),** *AI-Enhanced Stormwater Stewardship: Educating Communities to Protect Coastal Watersheds,* Mar 3, 2026.

Completed

- PCEC Undergraduate Research - Faculty Advisor, **Spring 2025** – Both teams won awards in the PCEC Research Symposium, 2025.
  - **(i) Multi-modal Data Analysis for Cognition, Emotion, and Attention Tracking, $1000 -** Anthony Irwin and Michael Landis - A pilot study to measure student engagement, performance, and emotional well-being using multimodal data analysis
  - **(ii) Designing Accountable Software Systems for Privacy Policy Transparency and Customer Rights Protection, $1000 -** Luke Blevins and Jessica Zumbach – Privacy policy analysis using graph mining, NLP, and graph mining.
- **HIPs Gen Ed Data Collection Project – PI, $10,000, 2022-2023,** UW-Parkside - To measure the *Impact of High Impact Practices (HIPs) in General Education (Gen Ed) and other courses* across the different disciplines (natural sciences, arts and humanities, and social sciences) at UW-Parkside
- **WiSys-Grant Writing Academy Fellow**, **$3000,** Spring/Summer 2022, UW-Parkside
- **University of Wisconsin-Parkside Online Course Development Grant, PI, $1000**, **2021.**
- **Undergraduate Research Apprenticeship Program (URAP), PI, $500, 2021,** UW-Parkside - Inspiring Women with Intent is an initiative to express the idea of creating a web-based mobile app to encourage women to pursue STEM/Computing
- **eLearning Research Grant - Innovative Learning Spaces, PI, $1000, 2019 – 2020,** Miami Univ. *Collaborative Network Modeling using Discussion Boards - A Student-Centered Learning Engagement Strategy*
- **CTE Major Teaching Research Project, PI, $1040, 2018 – 2019,** Miami University, *Analyzing Link Dynamics in Student Collaboration Networks using Canvas* to improve teaching-learning practices.
- **NSF DUE – 1525414, Investigator, $1.84M Grant, 2016 – 2019,** Miami University **https://stem-cyle.cis.fiu.edu/research-team** *Collaborative Research: Engaged Student Learning - Design and Development Level II: Using a Cyber Learning Environment to Improve Student Learning and Engagement in Software Courses,* Multi-Institutional Grant (FIU, NDSU, AAMU, FAMU, FGCU, GS, JSRCC, MU).
- **DSTO - UniSA - 2011/1167673/1, PI, $13000, 2012 – 2019**, Anna University, India, *Cognitive modeling and analysis of EEG data using Graph data model and efficient computational/ data mining algorithms.* Collaborative grant with CNeL – University of South Australia (UniSA).

Travel Grants totaling over $35,000 to present research at top-tier international conferences

Proposals in Preparation

- **NSF: RFE** Predicting and Preventing the Erosion of Engineering Identity to Support Students with ADHD - Co-PI – Due Mar 2026.
- **CAREER:** Student Progress and Retention through AI and Learning Assistant Coaching in Introductory Computer Science (SPARK) – PI – Due July 22, 2026.
- **NSF 25-543: CISE — Future CoRe (CHS/HCC)** Multimodal Sensing for Code Understanding: Privacy-Preserving Fusion of Gaze, Screen, and Audio for Real-Time Developer Support – Co-PI-Due Sep 10, 2026.

Proposals for Resubmission

- **Concept outline NSF Designing Accountable Software Systems (DASS) DCL: Project Title:** Collaborative: EAGER: FDASS: Exploratory Frameworks for Accountable Software Integrating Privacy Policy Transparency and Compliance, Performance, April 16, 2025
- **NSF S-STEM:** NSF Scholarships in Science, Technology, Engineering, and Mathematics Program (S-STEM) S-STEM: Empowering Academic Growth, Leadership, and Excellence in Computer Science (EAGLE-CS) ($2 M); PI: Viji Ramasamy. Provide scholarships and research experiences for academically talented yet financially challenged students in STEM disciplines - Due Mar 02, 2027.
- **Collaborative Research: DMREF: NSF-NSERC: Data-Driven Design and Fabrication of Imprinted Superabsorbent Polymers (ISAP) and their AI-based Binding Optimization to Pathogens (DMREF) (600K)** Co-PI.
- **NSF 25-026: Designing Accountable Software Systems(DASS)DCL:** *Exploratory Frameworks for Accountable Software Integrating, Privacy Policy Transparency and Compliance*, **Co-PI.**
- **NSF Scholarships in Science, Technology, Engineering, and Mathematics Program (S-STEM), PI, $2 M,** Feb 2026.
- **NSF Training-based Workforce Development for Advanced Cyberinfrastructure Training** ($1,000,000)**,** Evaluator: Viji Ramasamy, *CyberTraining: Spiral, Semantically Assisted Bilingual-Programming for Training HPC-CI Users and Contributors*
- **NSF DUE:** Collaborative Research: AI-INTEL: AI-Enhanced Online Discussion Boards for Engaged Learning – PI: Viji Ramasamy, $518,581 - The AI-INTEL (AI-Enhanced INTeraction Networks for Engaged Learning) project aims to transform computing education by integrating AI-based virtual assistants into a novel graph-based Interactive Discussion Board (IDB) model.

Research Impact

Scientific Journal Ranking Portal: http://www.scimagojr.com/journalrank.php

Title/URL Scopus Source ID	Impact Factor (SJR)	Scopus Sub-Subject Area	Publisher	# Articles
Neurocomputing	**5.779 Q1**	AI, CS Apps, Cognitive Neuroscience	Elsevier	**2**
Journal of Neural Engineering	**5.379 Q1**	Biomedical Engineering and Neuroscience	Institute of Physics Publishing (IOP)	**1**
Neural Processing Letters	**2.908 Q2**	AI, Com. Networks &Comm. Neuroscience	Springer Nature	**1**
Procedia Computer Science	**0.883**	Computer Science (all)	Elsevier	**2**
Journal of Graph Algorithms and Applications	**1.45 Q2**	Computational Theory and Mathematics	Brown University	**1**
Lecture Notes in Computer Science	**1.36**	CS, Theoretical CS	Springer Nature	**1**
Journal of Integrative Neuroscience	**2.7 Q2**	Neuroscience, Medicine (all)	Innovative Medical Research (IMR) Press	**4**
International Journal of Artificial Intelligence	**0.778**	AI	CESER Publications	**2**
Heliyon	**0.455 Q1**	Multidisciplinary	Elsevier BV	**1**

Journal of Engineering Education Transformations	**0.317 Q4**	Engineering, Social Sciences-Development Education	Rajarambapu Institute of Technology	**4**
Reviews in the Neurosciences	**4.1 Q1**	Behavioral, cognitive, and clinical neurosciences	Walter de Gruyter GmbH	**1**

Core Conference Ranking Portal: http://portal.core.edu.au/conf-ranks/

Title/DBLP/CORE Source	Acronym	Field of Research	Rank	# Art.
ACM Special Interest Group on Comp. Sci. Education Conference	**SIGCSE**	Human-centered computing	**A**	5
IEEE Conference on Frontiers in Education	**FIE**	Computing	**B**	5
International Conference on Computational Sciences	**ICCS**	Computation Theory and Mathematics	**A**	2
International Conference on Knowledge-Based and Intelligent Information and Engineering Systems	**KES**	Information Systems	**B**	1
International Conference on Neural Information Processing	**ICONIP**	Artificial Intelligence and Image Processing	**B**	1
International Conference on Pattern Recognition	**ICPR**	AI and Image Processing	**B**	1
Midwest Instruction and Computing Symposium	**MICS**	Computer Science in Higher Education	**-**	3
International Conference on Transformations in Engineering Education	**ICTEE**	Computer Science, Engineering Education	**-**	2
IEEE International Symposium on Software Reliability Engineering	**ISSRE**	Software engineering R&D	**A**	1
ACM Southeast (ACMSE) conference	**ACMSE**	Computer Science	**-**	3
International Conference on Availability, Reliability and Security	**ARES**	Dependability, Computer and Information Security	**B**	1
IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining	**ASONAM**	Data management and data science	**B**	2

Peer-Reviewed Publications (Rev. Chronological Order)[79 Refereed Publications]

Book/Book Chapters: 7; Conference: 49; Journal: 21; Workshop: 2

[a] Core Journal/Conference Ranking Portal: http://portal.core.edu.au/jnl-ranks

[b] Scientific Journal Ranking Portal: http://www.scimagojr.com/journalrank.php

*Current/Past Graduate Student co-authors; **Corresponding Author. 🎗 Awards;

Google Scholar Citations: 712; h-index: 14; i10-index: 20

In Progress

- Ragsdale, M*., and **Ramasamy, V**.,** "*AI-Driven Penetration Testing for ARM Systems: Comprehensive Experimental Evaluation and Deployment Framework Across Four AI Paradigms,*" to be submitted to IEEE Access by March 15, 2026.
- Ramasamy V., et al., "One Seminar, Five Campuses: A Shared Approach to Launching Learning Assistant Programs Across Computing Courses," 2026 ASEE Annual Conference & Exposition, 2026.
- Noushin Gauhar, Rushmila Shabneen, **Ramasamy, V.**, Md Shakil Hossain, and Andrew Allen, Comparative Analysis of Eye-metric Algorithms for Code Comprehension in an Introductory CS course, Technical Symposium on Computer Science Education
- Bardhan R., Griffith C., Dorai G., and **Ramasamy V.** *"PRACTICE: Perceptions of Digital Forensics Practitioners on AI and ML-Driven Data Analysis – A Comparative Evaluation."*

Submitted

- Ragsdale, M*., and **Ramasamy, V**.** "*Systematic literature analysis of artificial intelligence driven penetration testing for ARM systems,*" Submitted to Discover Artificial Intelligence, Springer, Feb 2026.

Accepted

2026

[ASEE 2026] [79] Clarke, P., Reid, M., **Ramasamy, V.**, Bogle, S.A., Narasareddygari, M., Fu, Y., and Jones, E. "*Experiences using Active Learning in Online Synchronous Learning Environments: A Multi-Institution Approach*," 2026 ASEE Annual Conference & Exposition, 2026. **[Core Rank: B]**

[AIRC 2026] [78] Raihan, N*., Ramasamy, V., Shabneen, R*., and Allen, A. "*Eye-Tracking Approaches to ADHD: A Comprehensive Scoping Review,"* Submitted to AIRC 2026 7th International Conference on Artificial Intelligence, Robotics and Control (AIRC), Savannah, GA, USA, 2025.

[77] Kaur, M*., **Ramasamy, V**.,** Allen, A. "*Evaluation Metrics in Knowledge Graph-Based Recommendation Systems: A Scoping Review,"* Accepted in AIRC 2026 7th International Conference on Artificial Intelligence, Robotics and Control (AIRC), Savannah, GA, USA, 2025.

[76] Hill, C., Jackson, J., Tong, W., Xu, Y., Li, L., and **Ramasamy, V.**, "*Sensor-Independent Wingbeat Spectrogram Dataset Construction and Hierarchical Insect Classification Using a CNN-Transformer Model,"* Accepted in AIRC 2026 7th International Conference on Artificial Intelligence, Robotics and Control (AIRC), Savannah, GA, USA, 2025.

Published

2026

[SICCSE 2026] [75] **Ramasamy, V**., Leshem, H., Reid, M., Tuttle, S., Solis, T., Ali, M., Jones, E., and Clarke. P.J., "*Investigating Student Belonging, Engagement, and Self-Efficacy in Online and In-Person Learning Environments,"* Technical Symposium on Computer Science Education (SIGCSE TS 2026), Feb 2026.901–907. https://doi.org/10.1145/3770762.3772626 **[Core Rank: A, Acceptance Rate: 30%]**

2025

[ASONAM-2025][74] Coggins, W., **Ramasamy, V.** (2026). COGRAM: A Computational Pipeline for Genome Assembly and Reconstruction via Optimized K-mer Sampling and De Bruijn Graph Networks. In: An, A., Cuzzocrea, A., Hu, H. (eds) Social Networks Analysis and Mining. ASONAM 2025. Lecture Notes in Computer Science, vol 16322. Springer, Cham. https://doi.org/10.1007/978-3-032-13513-1_31 **[Core Rank: B, Acceptance Rate: 25%]**

[73] Chowdhury, H., Morsalin, M.I., Azade, R.S., Ramasamy, V., Dorai, G. (2026). GraphDPR: A Privacy Policy Analysis Framework Using Knowledge Graphs and Topic Modeling. In: An, A., Cuzzocrea, A., Hu, H. (eds) Social Networks Analysis and Mining. ASONAM 2025. Lecture Notes in Computer Science, vol 16322. Springer, Cham. https://doi.org/10.1007/978-3-032-13513-1_33 **[Core Rank: B, Acceptance Rate: 25%]**

[EIT] [72] Ramamoorthy S., Sharma A., Anand S., and **Ramasamy V.,** Unveiling the generative power: An introduction to generative AI algorithms and their evolution. In Generative AI Unleashed (pp. 13–24) https://doi.org/10.1049/PBPC076E_ch2

[ICDF2C 2025] [71] Rajon Bardha, Gokila Dorai, Candace Griffith, **Ramasamy, V.**, *"Integration of NLP in Digital EAI Forensics: A Pilot Study of Practitioner Perceptions on Chat Data Analysis Tools,"* 16th EAI International Conference on Digital Forensics & Cyber Crime, 17-19 November 2025, Miami, Florida, USA **[CCF Rank: C, Acceptance Rate: 35%]**

[ARES 2025] [70] Gokila Dorai, Pouria Rad, Frank Breitinger, Rajon Bardhan, **Ramasamy V.** "*Mapping the Research Landscape - An Exploratory Analysis of AI Applications in Digital Forensics*," 20th International Conference on Availability, Reliability and Security (ARES 2025), Belgium, August 10-13, 2025. Lecture Notes in Computer Science, vol 15996. Springer, Cham. https://doi.org/10.1007/978-3-032-00635-6_7. **[[a]B, H5-Index: 27, Impact Score: 0.6]**

[ACMSE 2025] [69] Reddy M., **Ramasamy V.,** Cilone D., and Md Ali. *"Enhancing Computer Science Education using Learning Assistants: A Gamification Approach,"* 2025 ACM Southeast Conference (ACMSE 2025), Southeast Missouri State University, Cape Girardeau, Missouri. Apr. 24-26, pp. 171 – 179, 2025. https://doi.org/10.1145/3696673.3723075. **[Impact Score: 0.70]**

[IEEESC 2025] [68] Murry, B*., Kulpinski, E*., Aiello, A., **Ramasamy V**., Beaupre, T*., et al., *"Student Preferences in Interacting with AI-Enhanced Learning Assistants (AIELA): A Comparative Study,"*

IEEE SoutheastCon 2025, NC, March 28-29, pp. 501-507, 2025. doi:10.1109/SoutheastCon56624.2025.10971472. **[SJR: 0.205, Impact Score: 1.05]**

[IEEESC 2025] [67] S. Barrett, **V. Ramasamy,** G. Dorai, and B. Boswell, *"Navigating Privacy Policies with NLP and Graph Mining: Advancements in User-Centric Legal Document Analysis," IEEE SoutheastCon 2025*, Concord, NC, USA, pp. 1013-1022, 2025. doi: 10.1109/AIRC64931.2025.11077563. **[SJR: 0.205, Impact Score: 1.05]**

[AIRC 2025] [66] **V. Ramasamy,** S. Barett, G. Dorai, and J. Zumbach*, *"Unveiling Privacy Policy Complexity: An Exploratory Study Using Graph Mining, Machine Learning, and Natural Language Processing," 2025 6th International Conference on Artificial Intelligence, Robotics and Control (AIRC)*, Savannah, GA, USA, 2025, pp. 514-520, doi:10.1109/AIRC64931.2025.11077563.

[AIRC 2025] [65] R. Shabneen*, **Ramasamy, V.,** G. Shaver, M. S. Hossain, and A. Allen, "Optimizing Fixation and Saccade Classification for ADHD Assessment Using NiCATS and IVA-2," *2025 6th International Conference on Artificial Intelligence, Robotics and Control (AIRC)*, Savannah, GA, USA, 2025, pp. 128-132, doi:10.1109/AIRC64931.2025.11077514.

[Rev. Neurosci] [64] Manickam. T., **Ramasamy, V.,** and Doraisamy, N. *"Comparison of data-driven thresholding methods using directed functional brain networks,"* Reviews in the Neurosciences, De Gruyter (Walter de Gruyter GmbH), doi: 10.1515/revneuro-2024-0020. **[Impact Factor: 4.1, Q1, SJR: 1.213, H-Index: 88]**

2024

[SCERC 2024] [63] **Ramsamy V.,** *"Using EEG and Functional Brain Networks for Attention Detection."* **Southeast Community Engaged Research Conference (SCERC),** Georgia Southern University, USA.

[FIE 2024] [62] **Ramasamy, V.,** R Suganya, G Walia. *"Enhancing User Story Generation in Agile Software Development Through OpenAI and Prompt Engineering,"* IEEE Conference Frontiers in Education (FIE-2024), October 13-16, Washington-DC, USA. **[ªB, SJR: 1.213, H-Index: 49]**

[FIE 2024] [61] **Ramasamy, V.,** Eli Kulpinski, Thomas Beaupre, Aaron Antreassian, Yunhwan Jeong, Peter J. Clarke, Anthony Aiello, and Charles Ray. *"Enhancing CS Education With LAs Using AI-Empowered AIELA Program,"* IEEE Conference Frontiers in Education (FIE-2024), October 13-16, Washington, DC, USA. **[ªB, SJR: 1.213, H-Index: 49]**

[JEET 2024] [60] **Ramasamy, V.,** Ramamoorthy, S. "Fostering Student Engagement and Success in STEM
[ICTIEE 2024] Education: An AI Driven Exploration of High Impact Practices from Cross-Disciplinary General Education," **International Conference on Transformations in Engineering Education (ICTIEE 2024),** and Spl. Issue of Journal of Engineering Education Transformations (JEET), 2024. **[Impact Factor: 4.1 Q2, SJR: 0.317, H-Index: 15]**

2023

[JEET 2023] [59] **Ramasamy, V.,** Ramamoorthy, S. "Sustainable Data-Centric Project Curriculum for Industry
[ICTIEE 2022] Ready Workforce," (2023) Spl. Issue **Journal of Eng. Education Transformations (JEET)**, Vol. 36, pp. 99-105. D**OI**: 10.16920/jeet/2023/v36is2/23014. **[Impact Factor: 4.1 Q2, H-Index: 15]**

2022

[LAIS] [58] **Ramasamy V.**, Reddy, M., Walia, G., Kiper, J.D., Duke, D.M., Davis, D.L., Allen, A.A. "*A Multi-Institutional Analysis of CS1 Students' Common Misconceptions of Key Programming Concepts.*" Book Chapter, Learning and Analytics in Intelligent Systems, Vol. 3, pp 127–144, Springer, Cham. 2022. https://doi.org/10.1007/978-3-031-07650-3_8. **[Q4, SJR: 0.107, H-Index: 18]**

[FIE 2022] [57] Ojha H*., **Ramasamy V**., Walia G. *"Modeling Student Collaboration Network to Enhance Student Interactions,"* IEEE Conference on Frontiers in Education, FIE-2022, Oct. 8-11, Sweden. **[ªB, SJR: 1.213, H-Index: 49]**

[MICS 2022] [56] Chavan S*., Katare S*., Sutar S*., and **Ramasamy V**. *"Literature Review of Assessment of High Impact Practices in General Education Courses Using Text Mining and Natural Language*

Processing," **Midwest Instruction and Computing Symposium (MICS 2022),** Milwaukee School of Engineering (MSOE), USA, Apr 1, 2, 2022.

[MICS 2022] [55] Avachat S*., Dhawade A*., Mehta D*., and **Ramasamy V.** *"Car Accidents Severity Prediction in the USA Using Machine Learning Techniques,"* **Midwest Instruction and Computing Symposium (MICS 2022),** Milwaukee School of Engineering (MSOE), USA, Apr 1, 2, 2022.

[MICS 2022] [54] Rajput A*., **Ramasamy V.**, *"Machine learning approach for optimizing healthcare Supply chain."* **Midwest Instruction and Computing Symposium (MICS 2022),** Milwaukee School of Engineering (MSOE), USA, Apr 1, 2, 2022.

[JEET 2022] [53] **Ramasamy V.,** Suganya R., Sumesh K.V., *High Impact Practices & Collaborative Teaching*
[ICTIEE 2022] *to Enhance Learning and Engagement in Engineering Design Project Course,"* Special Issue of **Journal of Engineering Education Transformations**, Vol. 35, 2022. *eISSN-2394-1707,* **International Conference on Transformations in Engineering Education (ICTIEE 2022)**, Jan 7-9, 2022. **[Impact Factor: 4.1 Q2, SJR: 0.317, H-Index: 15]**

[AI Healthcare] [52] Kaushik. S., **Ramasamy, V***., Daniel, R., and Chandra, S. *"Characterizing EEG electrodes in directed functional brain networks using normalized transfer entropy and page rank," Book Chapter,* Handbook of Artificial Intelligence in Healthcare, Springer Nature Switzerland AG, pp 27-49, 2022.

2021

[ISSRE 2021] [51] Subedi, I., Singh, M., **Ramasamy, V**., and Walia. *"Classification of Testable and Valuable User Stories by using Supervised Machine Learning Classifiers,"* **IEEE International Symposium on Software Reliability Engineering - ISSRE 2021** Industry Track, China, Oct 25 - 28, 2021. **[ªA, H-Index: 46, Acceptance Rate: 30.3%]**

[Brain&BC] [50] Md. Hedayetul Islam Shovon*, D.(Nanda) Nandagopal, Jia Tina Du, and **Ramasamy. V,** "*Directed Functional Brain Networks Characterization of Information Flow Direction during Cognitive Function Using Non-Linear Granger Causality*,**" Brain and Behavior Computing**, pp. 357-378, Taylor & Francis, 2021.

[ACMSE 2021] [49] **Ramasamy, V**., Narasareddygari, M., Walia, G., et al., "*Meta-analysis to Study the Impact of Learning Engagement Strategies in Introductory Computer Programming Courses: A Multi-institutional Study,"* Proceedings of ACM Southeast (ACMSE) conference, pp 40–46, 2021.

[ACMSE 2021] [48] Desai, U., **Ramasamy, V.,** *"Evaluation of Student Collaboration on Canvas LMS Using Educational Data Mining Techniques,"* Proceedings of ACM Southeast (ACMSE) conference, pp 55–62, 2021.

[ACMSE 2021] [47] Subedi, I., Singh, M., **Ramasamy, V.,** and Walia. G. *"Application of Back-translation – A Transfer Learning Approach to Identify Ambiguous Software Requirements,"* Proceedings of The ACM Southeast (ACMSE) conference, pp 130-137, 2021.

2020

[HELIYON] [46] Alomari, H., **Ramasamy, V.,** Kiper, J.D., Potvin, G. "*A User Interface (UI) and User eXperience (UX) Evaluation Framework for Cyberlearning Environments in Computer Science and Software Engineering Education,*" Heliyon, Vol. 6, Issue 5, 2020. https://doi.org/10.1016/j.heliyon.2020.e03917 **[Impact Factor: 4.32 Q1, H-Index: 115]**

[SIGCSE] [45] Desai, U.*, ****Ramasamy, V.**, and Kiper, J. D. "*A Study on Student Performance Evaluation using Discussion Board Networks*," Proceedings ACM Special Interest Group on Computer Science Education (SIGCSE'20), Oregon, USA. **[ªA, Acceptance Rate = 32%, H Index: 43]**

2019

[FIE] [44] **Ramasamy, V.,** Kiper, J. D., Ojha. H*., Desai, U*. "*Analyzing Link Dynamics in Student Collaboration Networks using Canvas-A Student-Centered Learning Perspective,"* IEEE Conference on Frontiers in Education, FIE-2019, Oct. 16-19, OH, USA. **[ªB, Acceptance Rate = 56%, H Index: 35]**

[SIGCSE] [43] Reddy, M., *Walia, G.*, Duke, D., **Ramasamy, V**., Kiper, J., Davis, D., Allen, A., and Potvin, G. "*Evaluating the Impact of Combination of Engagement Strategies in SEP-CyLE to Improve Student Learning of Programming Concepts*", 50th ACM Technical Symposium on Computer Science Education. **[ªA, Acceptance Rate = 32%, H Index: 43]**

[WSDS] [42] **Ramasamy, V.**, Desai, U. "*Investigation of Learning Analytics and Data Visualization Techniques*," Women in Statistics and Data Science (WSDS-2019), USA

2018

[JIN] [41] Thilaga, M.,** **Ramasamy, V.**, Nadarajan, R., Nandagopal, D. "*Shortest Path-Based Network Analysis to Characterize Different Cognitive Load States of Human Brain using EEG based Functional Brain Networks*," Journal of Integrative Neuroscience, IOS Press, pp. 1-23, 2018 **[^{b}Q3, Impact Factor: 1.66, H Index: 29]**

[WSDS] [40] **Ramasamy, V.,** Miljkovic, T. *"Modeling Cyber Risk Using Network Theory,"* Women in Statistics and Data Science (WSDS-2018)

[WSDS] [39] Urvashi D*, **Ramasamy, V**. "*Content Analysis of Online Discussion Forums Using Predictive Analytics*," Women in Statistics and Data Science (WSDS-2018)

[ICC^3] [38] Thilaga, M., ****Ramasamy, V.**, Nadarajan, R., Nandagopal, D. "*Impact of Negative Correlations in Characterizing Cognitive Load States Using EEG Based Functional Brain Networks,"* Intl. Conference on Computational Intelligence, Cyber Security, and Comp. models (ICC^3-2018), *Springer*, pp. 74-86

[ICSCA] [37] **Ramasamy, V**., Desai, U., Alomari, H. W., and Kiper, J. D. "*TPGraphMiner: A Clustering Framework for Task-Based Information Networks,"* IEEE International Conference on Systems, Computation, Automation, Networking (ICSCA 2018), pp. 1-7

[SEEM] [36] **Ramasamy, V**., Alomari, H.W., Kiper, J.D., and Potvin, J. "*A Minimal Disruptive Approach of Integrating Testing into Computer Programming Courses*," 2nd IEEE/ACM Intl. Workshop on Software Eng. Education for Millennials (SEEM 2018), pp. 1-7

2017

[SERBIOENG] [35] Shovon M. H. I*., Nandagopal D., **Ramasamy, V**. "*Capturing Cognition via EEG Based Functional Brain Networks*," Emerging Trends in Neuro-Engineering and Neural Computation, Bio-Engineering book series, pp 147-172, *Springer.*

[PrCS] [34] **Ramasamy, V**., Nandagopal, D., Tran, M., and Abeynayake, C. "*A Novel Feature Extraction Algorithm for IED Detection from 2-D Images Using Minimum Connected Components*", Procedia Computer Science, Vol. 114, pp. 507-514 **[Impact Factor: 0.883, ^{b}Q2, H Index: 29]**

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[CCBR] [32] Thilaga M, **Ramasamy, V**., Nadarajan, R., Nandagopal, D. "*Challenges in Identifying Cognitive Activity using EEG based Functional Brain Networks*," 2nd CCBR workshop on Computational Brain Research, Indian Institute of Technology-IIT, Chennai, India

2016

[JIN] [31] Thilaga, M., ****Ramasamy, V**., Nadarajan, R., Nandagopal, D. A. "*Novel Pattern Mining Approach for Identifying Cognitive Activity in EEG-based Functional Brain Networks*," Journal of Integrative Neuroscience, Vol. 15(2), pp. 223-45 **[Impact Factor: 1.139, ^{b}Q3]**

2015

[ICC^3] [30] Senthilkumar, M, **Ramasamy, V**., Sheen, S., Veeramani, C., Bonato, A., Batten, L. (Eds.) Computational Intelligence, Cyber Security and Computational Models, Proceedings of 2nd International Conference on Computing, Computational Intelligence, Cyber Security and Computational Models, ISBN 978-981-10-0251-9, *Springer*

[SIST] [29] Nandagopal, D., **Ramasamy, V**., Cocks, B., Nabaraj, D., Dasari, N., Thilaga, M. "*Computational Neuroengineering Approaches to Characterizing Cognitive Activity in EEG Data*", Knowledge-Based Information Systems in Practice, Smart Innovation, Knowledge, Systems and Technologies, ISBN: 978-3-319-13544-1, pp. 115- 137, *Springer International Publishing Switzerland* **[^{a}B]**

[NeurIPS] [28] **Ramasamy, V**., Nandagopal, D., Thilaga, M., Cocks, B., **C.** "*Characterization of Cognitive Activity Using Minimum Connected Component*," Neural Information Processing, LNCS, ISBN: 978-3-319-26560-5, Vol. 9492, pp. 531-539 **[^{a}A]**

[NeurIPS] [27] Shovon, M.H.I*., Nandagopal, D., **Ramasamy, V**., Du, J.T., Cocks, B. "*Cognitive Load Driven Directed Information Flow in FBNs*," Neural Information Processing, Vol. 9492, pp 332-340 [[a]A]

[Neurocomputing] [26] **Ramasamy, V**., Nandagopal, D., Dasari, N., et al., "*Minimum Connected Component-A Novel Approach to Detection of Cognitive Load Induced Changes in Functional Brain Networks,"* Neurocomputing, Vol. 170, pp.15-31 [Impact Factor: 4.141, **[b]Q1]**

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[Neurocomputing] [25] Thilaga, M., **Ramasamy, V**., Nadarajan, R., Nandagopal, D., Cocks, B., Archana, C*., Dahal, N. "*A Heuristic Branch-and-Bound Based Thresholding Algorithm for Unveiling Cognitive Activity from EEG Data,"* Neurocomputing, Vol. 170, pp. 32-46 **[[b]Q1]**

[JIN] [24] Dasari, N., Nandagopal, D., **Ramasamy, V**., Cocks, B., Thomas, B., Nabaraj, D., Gaertner, P. "*Moment to moment Functional Brain Network variability during cognitive activity in EEG*", Journal of Integrative Neuroscience, Vol. 14, pp. 383-402 **[[b]Q3]**

[**SIGIR**] [23] Shovon M. H. I*., Nandagopal, D., Du J. T., **Ramasamy, V**., Cocks, B. "*Cognitive Activity during Web Search*," 38th ACM International Conference on Research and Development in Information Retrieval (ACM SIGIR 2015) **[[a]A, Acceptance Rate: 20%,]**

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[JNE] [22] Dahal, N., Nandagopal, D., Cocks, B., **Ramasamy, V**., Dasari, N., Gaertner, P, "*TVAR modeling of EEG to detect audio distraction during simulated driving,"* Journal of Neural Engineering, Vol. 11(3) **[[b]Q3, H Index: 89]**

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[JESP] [20] Cocks, B., Nandagopal, D., **Ramasamy, V**., Thilaga M, Dasari N, Dahal N. "*Brain Strain: How much is too much? Journal of Educational Sciences & Psychology,*" Vol. IV (LXVI), pp. 94-106

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[ICCS] [18] **Ramasamy, V**., Dasari, N., Nandagopal, D., Subhiksha, R*., et al., "*Change Detection and Visualization of Functional Brain Networks using EEG Data,*" International Conf. on Computational Sciences (ICCS-2014), pp. 672-682 **[[a]A, Acceptance rate: 30%]**

[ASE] [17] Shovon, M. H. I*., Nandagopal, D., **Ramasamy, V**., Du, J.T., Cocks, B. "*Towards a Cognitive Metric using Normalized Transfer Entropy*," ASE BigData/ SocialInformatics/ PASSAT/ BioMedCom 2014 Conference, Harvard University, pp. 1-9

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[PrSBS]
[ICCS] [16] Cocks, B., Nandagopal, D., **Ramasamy, V**., Thilaga, M., Dasari, M., Dahal, N. *"Breaking the Camel's Back: Can Cognitive Overload Be Quantified in The Human Brain?,"* Procedia – Social and Behavioral Sciences, Vol. 97(6) **Best Paper Award - 9th International Conference on Cognitive Science, Kuching, Malaysia, pp. 21-29 [Impact Factor 0.883, Acceptance rate: 26%]**

[PrCS] [15] Nandagopal, D., **Ramasamy, V**., Cocks, B., Dahal, N., Dasari, N., Thilaga, M. "*Computational Techniques for Characterizing Cognition using EEG - New Approaches*," Elsevier Journal of Procedia Computer Science, Vol.22, pp. 699–708 **[Impact Score; 2.09, [b]Q2]**

[AMS] [14] Nabaraj, D., Nandagopal, D., Cocks, B., Dasari, N., **Ramasamy, V**., Gaertner, P. "*Event-Related Synchronization & Desynchronization of EEG Data during Distracted Virtual Driving*," IEEE 7th Asia International Conference on Mathematical Modeling & Computer Simulation (AMS), pp. 23- 28 **[[a]C]**

[KES] [13] Nandagopal, D., **Ramasamy, V**., Cocks, B., Nabaraj, D., Dasari, N., Thilaga, M., Shamshu, D. S*. "*Computational Techniques for Characterizing Cognition using EEG - New Approaches*," 17th Intl. Conference on Knowledge-Based and Intelligent Information and Engineering Systems (KES-2013), pp. 699-708 **[Acceptance rate: 34%, [a]B]**

[ICCSII] [12] Dasari, N., Nandagopal, D., **Ramasamy, V**., Thomas, B., Cocks, B., Dahal, N., Gaertner, B. "*A Signal Processing System for Visualizing Complex Functional Brain Networks during Cognitive Load,"* IEEE ICCSII Conference, Indonesia

2012

[ACNS] [11] Dahal, N., Nandagopa,l D., Cocks, B., **Ramasamy, V**., Dasari, N. "*Modeling of Cognitive Functions in Driving Environments Using EEG*," 3rd Australasian Cognitive Neuroscience Conf. (ACNS 2012) **[Acceptance: 28.1%]**

[ACNS] [10] Dasari, N., Nandagopal, D., **Ramasamy, V**., Thomas, B., Dahal, N., Cocks, B. "*Visualization of Complex Neural connections during Cognitive load using EEG data*," 3rd Australasian Cognitive Neuroscience Conference (ACNS 2012), Australia [**Acceptance: 28.1%]**

[ICPR] [9] **Ramasamy, V**., Dahal, N., Dasari, N., Cocks, B., Nandagopal. D., "*Identification and Analysis of Functional Brain Networks*," International Conference on Pattern Recognition (ICPR 2012).

2011

[IJAI] [8] **Ramasamy, V**., Nadarajan, R., Thilaga, M. "*Performance Monitoring of Large Communication Networks using Maximum Common Subgraphs*," International Journal of Artificial Intelligence (IJAI), Vol. 6, pp.56-72 **[[b]Q2, H Index: 20]**

[IJAI] [7] **Ramasamy, V**., Nadarajan, R., Thilaga, M. "*A Divisive Clustering Algorithm for Performance Monitoring of Large Networks using Maximum Common Subgraphs*," International Journal of Artificial Intelligence, (IJAI), Vol. 7(11), pp. 92-109 **[[b]Q2, H Index: 20]**

[JGAA] [6] **Ramasamy, V**., Nadarajan, R., Thilaga, M., Nirmala, P. "*FPGraphMiner - A Fast Frequent Pattern Mining Algorithm for Network Graphs*," Journal of Graph Algorithms and Applications, (JGAA), Vol. 15(6), pp. 753 - 76 **[[b]Q2]**

2010

[IJOPCS] [5] **Ramasamy, V**., Nadarajan, R., Thilaga, M., Nirmala, P. "*A Novel Approach for Detection and Elimination of Automorphic Graphs in Graph Databases*," International Journal of Open Problems in Computer Science and Mathematics, Vol. 3(1), pp. 56-72

Before 2010

[ICIP] [4] **Ramasamy, V**., Nadarajan, R., Malar, B., "*A Study of Substructure Similarity Search in Graph Databases Using Grid-Based Approach*," Int. Conf. on Info. Proc. (ICIP-2007), 2007. [Acceptance rate: 25%], 2007

[NCDWDM] [3] **Ramasamy, V.** "*Data Warehousing*," National Conf. on Data Warehousing & Mining, 2003

[DDC] [2] **Ramasamy, V**. "*Multi-Agent System Engineering - A Methodology & Language for designing agent systems*," UGC Sponsored National Conference on Distributed Database & Computing, 2004

[NDDC] [1] **Ramasamy, V**. "*Conceptual Clustering of Heterogeneous Distributed Databases*," UGC Sponsored National Conference on Distributed Database and Computing, India, 2004

Selective Posters:

[1] Nasif Raihan, Christopher Page, Vijayalakshmi Ramasamy, and Andrew Allen. "Emerging AI-based Tools and Techniques for College-Level ADHD Detection: Integrating Eye-Tracking and Data-Driven Methods," MCG-GS Community Engaged Research Conference, Nov 17, 2025.

[2] Padmini Shankar, Vijayalakshmi Ramasamy. "AI-Driven Personalized Nutrition Strategies for Reducing Dietary Intake of Ultra-Processed Foods to Maintain Healthy Gut Microbiome," MCG-GS Community Engaged Research Conference, Nov 17, 2025.

[3] Collaborative Research: CUE-T: Broadening Participation in Computing via Active Learning Strategies, AAAS-IUSE, Collaborative Research: CUE-T: Broadening Participation in Computing via Active Learning Strategies - AAAS | IUSE

[4] Broadening Participation in Computing: A Nation-wide Online Live LA Pedagogy Course

Co-authors

Academia Florida International University, Augusta University, Georgia Southern University, Miami University, North Dakota State University, University of North Carolina, Virginia Commonwealth University, Rider University, Ryerson University, St. Cloud State University (USA); University of

South Australia, University of New England, Flinders University (Australia); Anna University (India)

Industry IBM India S/W Labs, Defence Science & Technology Org. Edinburgh-Australia, Institute of Nuclear Medicine and Allied Sciences, India, MiQ-Newyork, Dhuni Software-San Francisco Bay Area-USA

Summary of Synergetic Activities

Academic and Professional Service

Departmental Committees:

- *Graduate Committee Member*, CS Department, since Aug 2024
- *Graduate Thesis Committee advisor*: Jack N. Averitt College of Graduate Studies
  - ✓ Rushmila Shabneen, "*Enhancing ADHD Diagnosis in College Students Using Multimodal Integration of Nicats and Iva-2 Tools,"* Fall 2024-Spring 2025. https://digitalcommons.georgiasouthern.edu/etd/2950/
  - ✓ Noushin Gauhar, *"Comparative Analysis of Eye-Metric Algorithms for Code Comprehension in Introductory CS Course,"* Fall 2024-Spring 2025. https://digitalcommons.georgiasouthern.edu/etd/2946/
  - ✓ Lauren E. Perkins, *"Role of Eye-Tracking Technology and Software Algorithms in Enhancing ADHD Detection and Diagnosis: A Systematic Literature Review,"* https://digitalcommons.georgiasouthern.edu/honors-theses/1042/
- *Thesis Defense Committee Member: Advisor – Dr. Weitian Tong:*
  - Julia Jackson, Christopher Hill
- *Research Advising, Jack N. Averitt College of Graduate Studies, since Fall 2025*
  - ✓ *Advisor,* 3 Ph.D. Students: Pradhyum Nair, Oluwasola Smith, Mehak Kaur
  - ✓ *Co-Advisor,* 3 Master's Students: Christopher Page, Nasif Raihan
  - ✓ *Advisor,* 2 Master's Students: William Coggins, Matthew Ragdale
- **Capstone Projects Mentor [25 Undergraduate Students]**
  - ✓ 2025 – Privacy Policy Analysis Tool Project
  - ✓ 2024 - Discussion Board Analytics Project
  - ✓ 2022-24 - Canvas Analytics & NLP
  - ✓ 2022 – Canvas Analytics
  - ✓ 2018 – 2019 books2graphs
  - ✓ 2018 – 2019 Measuring Movement in the Brain

University Committes:

- Member - Faculty Senate's Ad Hoc AI Policy Committee – for shaping institutional policies on AI integration in education, Fall 2024 – Spring 2025.
- Alternate Faculty Senator – 2025 – 2026.

Professional Committes:

- Faculty Advisor, *Girls Who Code* Student Organization since Fall 2024.
- Institute for Health Logistics & Analytics (IHLA) *Affiliate* since Sep 2024.

NSF/Technical Programme Committee Member/Chair:

- *Review Committee Member Grant Proposal* - European Research Council (ERC) Advanced Grant, 2025
- *Member of National Science Foundation (NSF) Merit Review Panel*, Grant Proposals, Feb 2025.
- *TPC member***,** ICC3'25 - International Conference on Computational Intelligence, Cyber Security and Computational Models, Dec 2025.
- *Member of the Technical Program Committee***,** ACM Southeast (ACMSE) conference, 2025 ACM SE Conference, 2026 ACMSE Conference, Troy University, Troy, Alabama, April 23-25, 2026.
- *Senior PC member of Posters,* Technical Symposium on Computer Science Education (SIGCSE TS) 2026.
- *Publicity Chair,* 7th Int. Conference on Artificial Intelligence, Robotics, and Control (AIRC), Georgia Southern (Armstrong Campus), Savannah, May 7-9, 2025, April 8–10, 2026 AIRC 2026.
- *Session Chair:*
  - ✓ 7th Int. Conference on Artificial Intelligence, Robotics, and Control (AIRC), Georgia Southern (Armstrong Campus), Savannah, May 7-9, 2025, April 8–10, 2026 AIRC 2026.
  - ✓ 17th International Conference on Advances in Social Networks Analysis and Mining (ASONAM 2025), Canada, August 25-28, 2025.
  - ✓ IEEE Conf. Frontiers in Education (FIE-2024), Oct.13-16, Washington, DC, USA.
  - ✓ Machine Learning Track, Midwest Instruction and Computing Symposium (MICS 2022), Milwaukee School of Engineering (MSOE), USA, Apr 1, 2, 2022
  - ✓ Intl. Conf. on Systems, Computation, Automation, N/W (ICSCAN-2018), India.
- *International Program Committee Member:*
  - ✓ KES International Conf. on Innovation in Medicine & Healthcare (KES-InMed-24), June 2024, Portugal

- ✓ 10th KES International Conf. on Innovation in Medicine & Healthcare, 2023 (InMed-23), Greece
- ✓ 6th Intl. Conf. on Computational Intelligence, Cybersecurity & Computational Models (ICC^3 2023), India

M.S. Thesis Advisor: Miami University:

- 2017-20 Urvashi Desai, MS Computer Science; Thesis: *Plug-In for Canvas LMS that allows users to visualize and grade the network of interactions resulting from discussion forum posts in the course Canvas using social network metrics* (Developer at World Fuel Services, USA), Link: MS Thesis advisor_Urvashi-2020
- 2018-20 Hemraj Ojha, MS Computer Science; Thesis: *Analyzing link dynamics in collaboration networks using schema-based structured network Models* (Software Engineer at LIS Nepal Pvt. Ltd, Nepal), Link: MS Thesis advisor_Hemraj-2020

Ph.D. Associate Supervisor: University of South Australia:

- Md. Hedayetul Islam Shovon, School of Information Technology and Mathematical Science, UniSA-Dissertation: *Identification of Cognitive Activity using Multidimensional EEG Database during Cognitive Load, including Multitasking Web Search* (Specialist/Scientist at Defence Science and Technology (DST), Australia)

Reviewer – International Journals:

- Handbook of AI in Healthcare-Springer, Scientific Reports-Nature, Journal of Pattern Recognition Letters-Elsevier, Journal of Information Systems-Elsevier, International Journal of Knowledge Engineering & Soft Data Paradigms, Journal of Integrative Neuroscience (JIN)-IOS Press, IEEE Transactions on Education, IEEE Transactions on Learning Technologies, TLT-2025-06-0467, Journal of Software and Systems Modeling (SoSyM)-Springer

Reviewer – Peer-Reviewed Conferences:

- IEEE Global Engineering Education Conference (EDUCON 2025)**,** International Conference on Computational Intelligence, Cyber Security and Computational Models (ICC3'25) 2025., Technical Symposium on Computer Science Education (SIGCSE TS) 2026, *SERA 2025 special session: Transformative Technologies: Big Data, Machine Learning, Cybersecurity, and Blockchain for the Next Era of Digital Innovation.* SERA 2025 Special Sessions – ACIS International, IEEE Conference of Frontiers in Education (FIE), ACM Symposium on Computer Science Education (SIGCSE), 7th Edition of International Conference on Computational Intelligence Cyber Security and Computational Models (ICC3 2025), December 18-20, 2025

Honors and Awards:

- *Best Paper of the Session Award* - Artificial Intelligence, Robotics and Control (AIRC) 2025
- *Faculty Advisor, Undergraduate Research* - Student Projects – Advisees won first and second places in the PCEC Research Symposium, 2025
- *Advisor, "Discussion Board Analytics Project," Capstone Project Fall 2024, Spring 2025,* Georgia Southern University
- *Chair,* Computer Science, May 2023-July 2024, UW-Parkside
- *Academic Director,* Master of Science Computer Information Systems (MSCIS) program, May 2023-July 2024, UW-Parkside
- MoU renewal, *international transfer agreement with MIT-World Peace University* to increase enrollment, admit, and advise graduate students, UW-Parkside
- *Learning Assistant (LA) program* in Data Structures and Algorithm Design and CS1 courses, Spring 2024
- *Graduate Research Advisory Committee,* Vishwanath Karad MIT World Peace University, India, PhD scholar Mr. Chinmay Patwardhan, from Sep 2025
- FYE 1220 SAH - Computer Science Guest Lecture @ Wed Sep 10, 2025 4 pm - 4:50 pm (EDT), Research Presentation, CSCI 7630 Current Trends in Computing, Oct 31, 2025
- Professional Development Day, AEP College of Engineering and Computing (AEP COEC), Seminar #1---Collaborative Proposals, Jan 2025
- Faculty Center - Learning that Matters - Fall 2024
- VIP program Mentor – Summer and Fall 2025
- Delivered a *special lecture* on *"Artificial Intelligence and Technology: Opportunities & Impacts"* to FYE-VIP 1211 Introduction to Research course, Oct 28, 2025.
- *WiSys-Grant Writing Academy Fellow*, UW-Parkside, USA
  - ✓ Fall 2023, (Link)
  - ✓ Spring/Summer 2022 (Link)
- *Teaching Excellence in Computer Science Award* Nominee, UW-Parkside, 2021-2023.
- *Advisory board Member,* College of Business, Economics, and Computing (CBEC), UW-Parkside, 2021-2023.

- *Best Research Paper Award***,** International Conference on Transformations in Engineering Education (ICTIEE-2022), India, published in the special issue of ***Journal of Engineering Education Transformations***, Vol. 35, 2022
- WiSys Quick Pitch Competition, UW-Parkside, 2021 – Student advisees won First and Second places and the People's Choice award.
- *Research Advisory Committee,* Ph.D. Computer Science Program, School of Computer Science, MIT World Peace University, India, since 2021
- Coordinator - *CSE 174,* 2017-18 - *CSE Committee, for* 12 sections of ~300 students
- *Graduate Thesis Defense Committee Member* (2017-2020)– External examiner for FOUR M.S. Thesis Candidates
- Offered Independent Study for Graduate Research Program, Miami University
- *Investigator* - NSF Research Grant: NSF DUE – 1525414: Collaborative Research, published research
- Member of the Board of Studies. M.Sc Software Engineering, M.Sc Theoretical Computer Science, and M.Sc Data Science programs – 2008 - 2016
- Member Graduate and Undergraduate Advising Committee 2008 – 2016
- Expert Member On-Site review to evaluate internship projects of graduate students at Software Companies IBM, Yahoo, Amazon, Oracle, TCS, CTS 2008 – 2016
- Course-Coordinator Programming courses (C Programming, Object Oriented Programming with C++) 18 Undergraduate Engineering Majors 2005-15 AU, India
- Level B standing Graduate Faculty at the Department of Computer Science and Software Engineering, 2017.
- *Best Paper Award*, 9th International Conference on Cognitive Science, Malaysia, 2013.
- *Outstanding Faculty Research Award*, Anna University, India, 2013-2016.
- *Best Outgoing Student*, M.C.A., 1996, B.Sc. Chemistry, 1993, Bharathiar University, India.
- *University II Rank*, B.Sc. Chemistry, Bharathiar University, India, 1993.

Certifications

- *UW System Certificate in Online Learning*, UW-System, June 2021
- *Applied Deep Learning, Computer Vision & NLP with Python* – IIT Hyderabad 30 Hrs -Feb 2021
- *Information Security Awareness Training - UW-Parkside* – 2021
- *Code of Conduct: Data Security (US; Multi Lang)* - UW-Parkside – 2021
- *Harassment & Discrimination Prevention* – UW-Parkside – 2021
- *University of Wisconsin System Certificate in Online Learning* – UW-Parkside, Dec 2020
- *Managing Bias Certificate* - Miami University - 2020
- *Diversity: Inclusion in the Modern Workplace Certificate* - Miami University – 2020
- *Human Research (Curriculum Group) - Humans as Subjects IRB Basic Course* – 2018
- *Graph Analytics for Big Data* - US Sandiago – 2017
- *A Crash Course in Data Science* - John Hopkins University - 2016
- *Internal Auditor Training* - Total Quality Management Intl. Pvt.Ltd., (TQMI) - 1997
- *Diploma in Database Management (DDBM)* - Info-Drive Software Ltd - *1996*

Keynotes/ Special Lectures

- *Assessment of High Impact Practices in Education using Machine Learning Techniques,* Invited research talk - Wisys SPARK-2022 Symposium, UW-La Crosse, Jul 31-Aug. 2, 2022
- *Engineering Design Project for improving Engineering Exploration, and Lateral, Design and System thinking* - Webinar Series - Thiagarajar College of Engineering, India, 2021
- *Careers Involving Quantitative Skills (CIQS) Day*, MU, Jan. 2017, 2018, 2019
- *Functional Brain Networks*, Department of Psychology, Miami University, Sep. 2016
- *Symposium on Cognitive Neuro-Eng. & Computational Neuroscience*, UniSA, Jul 2013, Feb. 2015
- *Brain Network Analysis – Computational Neuroscience* - ACRC-Research Seminar series, UniSA, July 2012
- *Graph Mining*, Cognitive-Neuro Engineering Laboratory (CNEL), UniSA, July 2012
- *Curriculum Development Committee* – AI Center for Excellence, 2022-24
- *Undergraduate Program Task Force Committee* – Reduce Time and Cost for Undergraduate Education, 2022-24
- *Business Data Analytics Programme Committee* to propose a new Data Analytics program - Member and PI, 2023-24
- *Search & Screen Committee* Lecturer of Computer Science – Spring 2022 and Summer 2022
- *Faculty Search & Screen Committee* Assistant/Associate Professor of Computer Science – Spring 2023
- *Information Technology Practice Center (ITPC)* - Member

- *Institutional Review Boards (IRB) Committee* - Member
- *General Education Committee* – Member
- *Started Girls Who Code- College Loops Club in Oct 2023,* advisor
- Active member of the **Learning Assistants (LA) program**, attended the following conferences and workshops:
- *International Learning Assistant Conference (ILAC 2023), Boulder, Colorado Oct 2023*
- *LAA eSeries workshops organized by the Learning Assistant Alliance (LAA)* Oct 2023:
  - ✓ Transforming Your Course with LAs Workshop
  - ✓ Essential Elements of the LA Model
- *MS CIS Student Project Advisor* in CIS 690 Data Science/Machine Learning Project course:
- Presented papers at the Midwest Instruction and Computing Symposium (MICS) on April 1-2, 2022.
- Presented posters at the UW-Parkside Student Showcase on April 20, 2022.
- Highlights of WiSys Quick Pitch Competition held at UW-Parkside - Student Advisees won first and second places and the People's Choice award.
- *3-day BRIDGES Workshop (May 23-25, 2022) - Stipend $3000 - University of North Carolina (UNC)-"Real World Data, Visualizations, and Engaging Assignments Using BRIDGES*
- *Supervisor New Student Orientation Training* 2021
- *Online Course Developers Workshop*, June 2021
- *Women in Statistics and Data Science Workshop – (WSDS 2021).*
- *Summer Internship Workshop*, June-July 2021
- *High Impact Practices in General Education*, June 2021
- *Summer Institute: Creating an Anti-Racist Classroom Through Teaching & Learning*, June 2021
- *Common Grant Writing Pitfalls* Nov. 5, 2020
- *Tools for Finding Funding: PIVOT/Grants Resource Center* Oct. 1, 2020
- *New Faculty Colloquium: Classroom Assessment Techniques, High Impact Practices* (6 Mon.) 2020